

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

|                             |                                                       |
|-----------------------------|-------------------------------------------------------|
| <b>Perihalan Produk:</b>    | <b>Potassium hydrogen phthalate, primary standard</b> |
| <b>Product Description:</b> | <b>Potassium hydrogen phthalate, primary standard</b> |
| <b>Cat No. :</b>            | 33351                                                 |
| <b>Synonyms</b>             | Potassium acid phthalate; Potassium biphthalate       |
| <b>CAS No</b>               | 877-24-7                                              |
| <b>Molecular Formula</b>    | C8 H5 K O4                                            |

**Relevant identified uses of the substance or mixture and uses advised against**

|                             |                          |
|-----------------------------|--------------------------|
| <b>Recommended Use</b>      | Laboratory chemicals.    |
| <b>Uses advised against</b> | No Information available |

|                |                                                                                                                                                                                                                                                       |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Company</b> | Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd<br>Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,<br>No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,<br>Selangor Darul Ehsan, Malaysia.<br>Main line: +60 3-5525 7888 |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Supplier**

|                       |                             |
|-----------------------|-----------------------------|
| <b>E-mail address</b> | Enquiry.my@thermofisher.com |
|-----------------------|-----------------------------|

|                                   |                                                                                                                                         |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Emergency Telephone Number</b> | Tel: +03-5525 7888<br>CHEMTREC Malaysia <b>1-800-815-308</b> (Malay)<br>CHEMTREC Malaysia (Kuala Lumpur) <b>+(60)-327884561</b> (Malay) |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture****Label Elements**
**Hazard Statements**
**Precautionary Statements**
**Prevention**

P270 - Do not eat, drink or smoke when using this product

**Response**

P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell

**Storage**

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P403 - Store in a well-ventilated place

## Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                                        | CAS No   | Weight % |
|--------------------------------------------------|----------|----------|
| 1,2-Benzenedicarboxylic acid, monopotassium salt | 877-24-7 | >95      |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention if irritation persists.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

#### Ingestion

Do NOT induce vomiting. Get medical attention.

#### Inhalation

Remove to fresh air. Get medical attention immediately if symptoms occur. If not breathing, give artificial respiration.

#### Self-Protection of the First Aider

No special precautions required.

### Most important symptoms and effects, both acute and delayed

No information available.

### Indication of any immediate medical attention and special treatment needed

#### Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### Extinguishing media which must not be used for safety reasons

No information available.

### Special hazards arising from the substance or mixture

Keep product and empty container away from heat and sources of ignition. Thermal decomposition can lead to release of irritating gases and vapors.

### Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

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## Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

### Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Avoid dust formation.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

### Exposure Controls

#### Engineering Measures

None under normal use conditions.

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles)

#### Hand Protection

Protective gloves

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

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(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

## Respiratory Protection

### Recommended Filter type:

No protective equipment is needed under normal use conditions

Particle filter

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

No special environmental precautions required

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                                  |                             |                                   |
|--------------------------------------------------|-----------------------------|-----------------------------------|
| Appearance                                       | White                       |                                   |
| Physical State                                   | Solid                       |                                   |
| Odor                                             | Odorless                    |                                   |
| Odor Threshold                                   | No data available           |                                   |
| pH                                               | 3.8-4.0                     | 5% aq.sol                         |
| Melting Point/Range                              | 295 - 300 °C / 563 - 572 °F |                                   |
| Softening Point                                  | No data available           |                                   |
| Boiling Point/Range                              | No information available    |                                   |
| Flash Point                                      | No information available    | Method - No information available |
| Evaporation Rate                                 | Not applicable              | Solid                             |
| Flammability (solid,gas)                         | No information available    |                                   |
| Explosion Limits                                 | No data available           |                                   |
| Vapor Pressure                                   | No data available           |                                   |
| Vapor Density                                    | Not applicable              | Solid                             |
| Specific Gravity / Density                       | No data available           |                                   |
| Bulk Density                                     | No data available           |                                   |
| Water Solubility                                 | 80 g/L (20°C)               |                                   |
| Solubility in other solvents                     | No information available    |                                   |
| Partition Coefficient (n-octanol/water)          |                             |                                   |
| Component                                        | log Pow                     |                                   |
| 1,2-Benzenedicarboxylic acid, monopotassium salt | <=-3.9                      |                                   |
| Autoignition Temperature                         | No data available           |                                   |
| Decomposition Temperature                        | No data available           |                                   |
| Viscosity                                        | Not applicable              | Solid                             |
| Explosive Properties                             | No information available    |                                   |
| Oxidizing Properties                             | No information available    |                                   |

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Molecular Formula C8 H5 K O4  
Molecular Weight 204.22

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

**Hazardous Polymerization** Hazardous polymerization does not occur.  
**Hazardous Reactions** None under normal processing.

### Conditions to Avoid

Incompatible products. Excess heat. Avoid dust formation.

### Incompatible Materials

Strong oxidizing agents.

### Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

**Oral** Based on available data, the classification criteria are not met  
**Dermal** No data available  
**Inhalation** No data available

| Component                                        | LD50 Oral                 | LD50 Dermal | LC50 Inhalation |
|--------------------------------------------------|---------------------------|-------------|-----------------|
| 1,2-Benzenedicarboxylic acid, monopotassium salt | LD50 > 3200 mg/kg ( Rat ) | >1000 mg/kg | -               |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

##### (d) respiratory or skin sensitization;

**Respiratory** No data available  
**Skin** No data available

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|                             |                                                                                |
|-----------------------------|--------------------------------------------------------------------------------|
| (e) germ cell mutagenicity; | No data available                                                              |
| (f) carcinogenicity;        | No data available<br>There are no known carcinogenic chemicals in this product |
| (g) reproductive toxicity;  | No data available                                                              |
| (h) STOT-single exposure;   | No data available                                                              |
| (i) STOT-repeated exposure; | No data available                                                              |
| Target Organs               | No information available.                                                      |
| (j) aspiration hazard;      | Not applicable<br>Solid                                                        |

Symptoms / effects, both acute and delayed No information available.

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects** Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

**Persistence and degradability** Readily biodegradable  
**Persistence** Soluble in water, Persistence is unlikely, based on information available.

**Bioaccumulative potential** Bioaccumulation is unlikely

| Component                                        | log Pow | Bioconcentration factor (BCF) |
|--------------------------------------------------|---------|-------------------------------|
| 1,2-Benzenedicarboxylic acid, monopotassium salt | <=-3.9  | No data available             |

**Mobility in soil** The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

**Waste treatment methods**

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## Waste from Residues/Unused Products

Dispose of in accordance with local regulations

## Contaminated Packaging

Empty containers should be taken to an approved waste handling site for recycling or disposal

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

Not regulated

### Road and Rail Transport

Not regulated

### IATA

Not regulated

### Special Precautions for User

No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

X = listed

| Component                                        | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|--------------------------------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| 1,2-Benzenedicarboxylic acid, monopotassium salt | 212-889-4 | X    | X   | X     | X    | X    | X     | X    | KE-02310 |

### National Regulations

#### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

POW - Partition coefficient Octanol:Water

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

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**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Prepared By**

Health, Safety and Environmental Department

**Revision Date**

24-Mar-2025

**Revision Summary**

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**